

MT 301: Phase Transformations

Course Code	MT 301
Course Name	Phase Transformations
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech. (MSE)
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Module-1:** Overview of phase transformations, phase equilibria, thermodynamics of phase changes, free energy, order of transformation, thermodynamic driving force; nucleation and growth theories, homogeneous and heterogeneous nucleation, growth kinetics, precipitation and phase separation - spinodal microstructures, crystal interfaces and microstructure. (6 Hours)
- **Module-2:** Eutectic and peritectic transformation, heat-treatment of ferrous and nonferrous metals, eutectoid, phase diagrams and equilibria in relation to free energy-composition diagrams. interpretation of phase diagrams, determination and calculations, solid-liquid miscibility gap; invariant reaction. Principles of ternary phase diagram, examples of a few metallic and ceramic phase diagrams. (12 Hours)
- **Module-3:** Principles of Solidification in metals and alloys: thermodynamics involved, eutectic and peritectic Solidification, Homogeneous and heterogeneous nucleation, Mechanisms of growth. Rapid Solidification Processing. Austenite, transformation of austenite, TTT diagram, eutectoid transformation, pearlite and bainite transformation, order-disorder transformation, precipitation hardening, spinodal decomposition and massive transformation, solidification. (8 Hours)
- **Module-4:** Martensite transformation: characteristics and nature, morphology, crystallography, theory of nucleation and growth, and pre-martensite phenomena, martensitic transformation in steel. Martensite in non-Ferrous systems-thermoelastic martensite and shape memory effect (8 Hours)
- **Applications:** Smart materials and actuators, shape memory effect, metal-insulator transition. (4 Hours)

Laboratory Modules

1. NA

Text Books

1. D.A. Potter and K.E. Easterling, **Phase transformations in metals and alloys**, CRC Press, 1992.
2. P.G. Shewmon, **Transformations in Metals**, Mc-Graw Hill, 1969.
3. S. N. Avner, **Introduction to Physical Metallurgy**, Tata McGraw Hill, 1997.
4. Peter Haasen, **Physical Metallurgy**, Cambridge University Press, 1996.

References

1. A. G. Guy, **Physical Metallurgy for Engineers**, Addison-Wesley Pub. Co., 1962.
2. R. E. Smallman, **Modern Physical Metallurgy**, Butterworths, 1963.